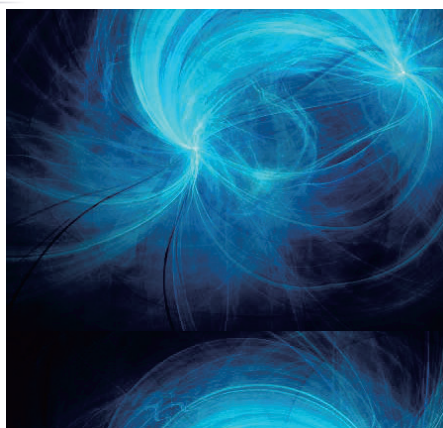


Unit 5: Electromagnetism

AP Physics C Electricity and Magnetism



Contents

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01 Electromagnetic Induction

02 Inductance

03 RL and LC Circuits

04 Maxwell's Equations

TESTDAILY

Electromagnetic Induction

5.1

TESTDAILY

Electromagnetic Induction

5.1.1 — Motional EMF

5.1.4 — Lenz's Law

5.1.2 — Magnetic Flux

5.1.5 — Induced Electric Fields

5.1.3 — Faraday's Law

TESTDAILY

5.1 Electromagnetic Induction

5.1.1 Motional EMF

5.1.1

Concept

TESTDAILY

Motional EMF

- Perpendicular wire, velocity, and magnetic field – **cutting** magnetic field lines
- Magnetic force on electrons
- Internal electric field
- Equilibrium

$$qvB = qE$$

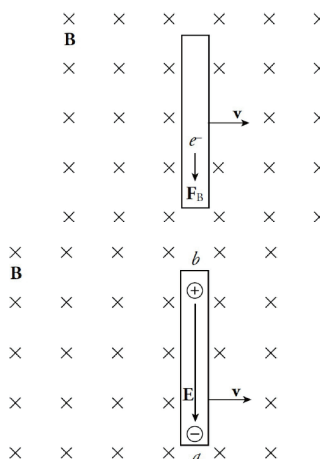
- Potential difference and induced EMF

$$\varepsilon = \Delta V = Blv$$

- Determine the end with higher potential

$$+ : \vec{v} \times \vec{B}$$

5.1.1

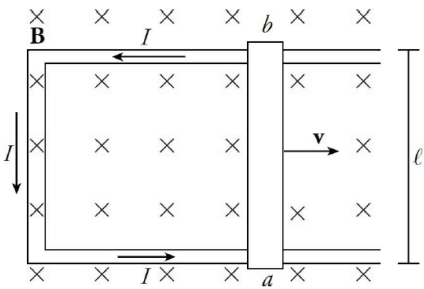


Property

TESTDAILY

Motional EMF

- Induced current in the loop
- Magnetic force on current
- Newton’s second law and differential equation

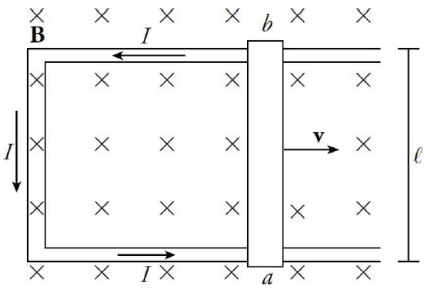


Property

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Motional EMF

- Electrical power dissipated in circuit
- Power supplied by external force



5.1 Electromagnetic Induction

TESTDAILY

5.1.2 Magnetic Flux



Introduction

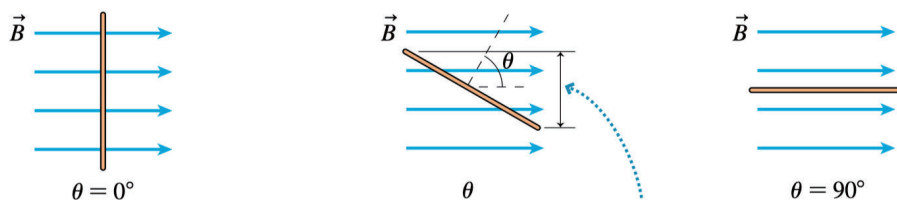
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Magnetic Flux

Similar to electric flux, for uniform field and flat surface:

$$\Phi_B = BA \cos \theta = B_{\perp} A = \vec{B} \cdot \vec{A}$$

Loop seen from the side:



Concept

TESTDAILY

Magnetic Flux

- Divide any surface into elements of area dA .

Determine the perpendicular component $B_{\perp}$ of magnetic field

$$B_{\perp} = B \cos \theta.$$

- The magnetic flux through this area is

$$d\Phi_B = B_{\perp} dA = B \cos \theta dA = \vec{B} \cdot d\vec{A}$$

- The total magnetic flux through the surface is the sum of the contributions from the individual area elements.

Physical quantity	Magnetic flux 磁通量
Definition	The scalar product of the magnetic field and the area vector over the entire area
Symbol	Φ_B
Equation	$\Phi_B = \int \vec{B} \cdot d\vec{A}$
Unit	weber, Wb $1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$



$$\Phi_B = \int d\Phi_B = \int \vec{B} \cdot d\vec{A}$$

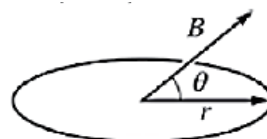
Exercise

TESTDAILY

2013_Intl_MCQ_6

A circular loop of radius r is located in a uniform magnetic field of magnitude B directed at an angle θ to the plane of the loop, as shown above. What is the magnetic flux through the loop?

- (A) $\pi r^2 B \sin \theta$
- (B) $\pi r^2 B \cos \theta$
- (C) $\pi r^2 B$
- (D) $2\pi r B \cos \theta$
- (E) $2\pi r B$



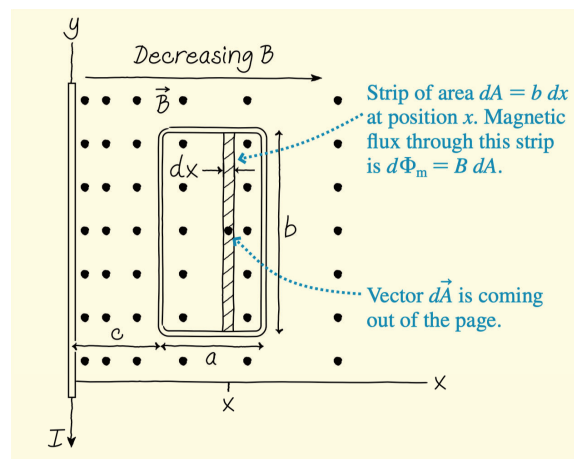
Example

TESTDAILY

Magnetic Flux from the Current in a Long, Straight Wire

The rectangular loop has dimensions a and b , as shown, with the near edge at distance c from the wire.. The wire carries a current of I . What is the magnetic flux through the loop?

$$\Phi_B = \frac{\mu_0 I b}{2\pi} \ln\left(\frac{c+a}{c}\right)$$



5.1.2

5.1 Electromagnetic Induction

5.1.3 Faraday's Law

5.1.3

Concept

TESTDAILY

Faraday's Law (CED p95)

The induced emf in a closed loop equals time rate of change of magnetic flux through the loop.

$$\varepsilon_i = -N \frac{d\Phi_B}{dt}$$

$$|\varepsilon_i|_{\text{avg}} = N \left| \frac{\Delta\Phi_B}{\Delta t} \right|$$

$$|\varepsilon_i|_{\text{inst}} = N \left| \frac{d\Phi_B}{dt} \right|$$

5.1.3

Concept

TESTDAILY

Faraday's Law

- What can cause the flux to change?
 - Changing area
 - Changing magnetic field
 - Changing orientation

5.1.3

Concept

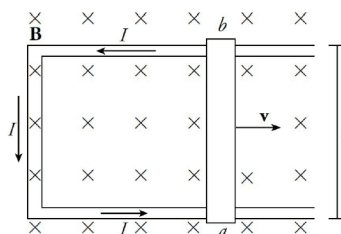
TESTDAILY

Faraday's Law

Case I: Changing area

- Motional emf revisited

$$|\varepsilon_i| = N \left| \frac{d\Phi}{dt} \right| = NB \left| \frac{dA}{dt} \right|$$



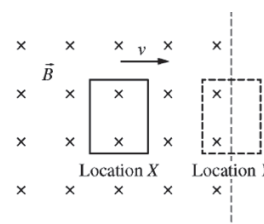
5.1.3

Exercise

TESTDAILY

2016_Intl_MCQ_22

A loop of wire at location X is moving toward the right with constant speed v in a region of uniform magnetic field $\vec{B}$, which is perpendicular to the plane of the loop. The magnetic field region ends at the dashed line, as shown in the figure above. Later, the loop is at location Y and exiting the magnetic field with the same constant speed v . The process is then repeated with the loop moving at a speed of $2v$. Which of the following best describes the emf in the loop at the two positions shown when the process is repeated at a speed of $2v$?

Emf at location X Emf at location Y

- | | |
|-------------------------|---------------------|
| (A) Nonzero and halved | Nonzero and halved |
| (B) Nonzero and doubled | Nonzero and doubled |
| (C) Nonzero and doubled | Zero at both speeds |
| (D) Zero at both speeds | Zero at both speeds |
| (E) Zero at both speeds | Nonzero and doubled |

5.1.3

Concept

TESTDAILY

Faraday's Law

Case II: Changing magnetic field

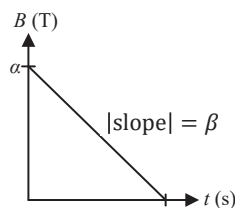
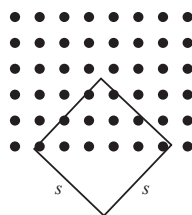
$$|\varepsilon_i| = N \left| \frac{d\Phi}{dt} \right| = NA \left| \frac{dB}{dt} \right|$$



Exercise

TESTDAILY

2021_DE_D61102_11



A square loop with sides of length s is perpendicular to a uniform time-varying magnetic field with half the area of the loop in the field, as shown. The loop resistance is R . The magnitude of the field varies with time according to the graph, where β is the absolute value of the slope and α is the value of the vertical intercept. The magnitude of the induced current in the loop is

- A. $\frac{\alpha s^2}{2R}$ B. $\frac{\alpha s^2}{R}$ C. Zero D. $\frac{\beta s^2}{R}$ E. $\frac{\beta s^2}{2R}$



Concept

TESTDAILY

Faraday's Law

Case III: Changing orientation

$$|\varepsilon_i| = N \left| \frac{d\Phi}{dt} \right| = NBA \left| \frac{d(\cos \theta)}{dt} \right| = NBA\omega |\sin \theta|$$

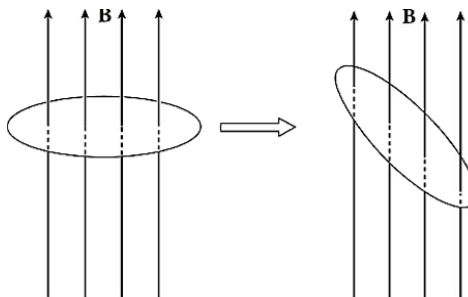


Example

Princeton_2020_Chapter16_Example1&2

The circular loop of radius 3 cm is placed within a uniform magnetic field, B (magnitude 0.2 T) and rotates at a constant angular speed through 45° in 0.5 s.

- What's the magnetic flux through the loop at the two positions?
- What's the average induced emf in the loop?
- What's the instantaneous induced emf in the loop?



Answer: 5.7×10^{-4} Wb, 4.0×10^{-4} Wb, 3.4×10^{-4} V, 6.3×10^{-4} V.

5.1.3

5.1 Electromagnetic Induction

5.1.4 Lenz's Law

5.1.4

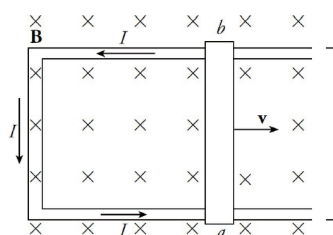
Concept

Lenz's Law (Young 13e p967)

- Direction—Lenz's law

Expression 1:

The direction of any magnetic induction effect is such as to oppose the cause of the effect.



5.1.4

Concept

TESTDAILY

Lenz's Law (CED p95)

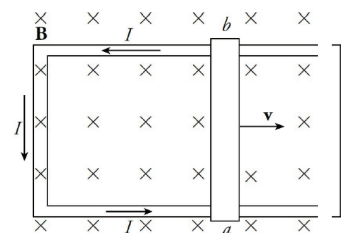
- Direction—Lenz's law

Expression 2:

The induced EMF and current induced in a conductive loop will create an induced magnetic field to oppose the change in external flux.

Three steps

1. Change in flux?
2. Induced magnetic field?
3. Induced current?

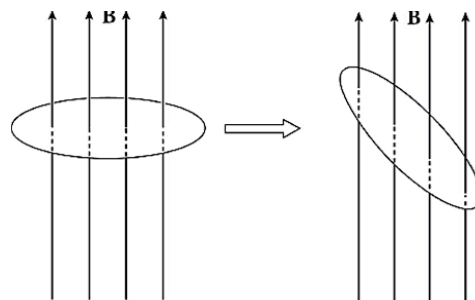


Example

TESTDAILY

Princeton_2020_Chapter16_Example2

In which direction will current be induced to flow?



Answer: in the counterclockwise direction (looking down on the loop).



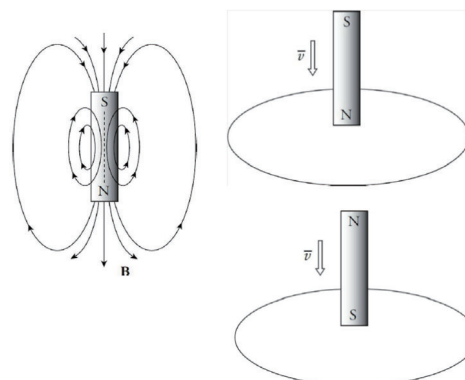
Example

TESTDAILY

Princeton_2020_Chapter16_Example4

A permanent magnet creates a magnetic field in the surrounding space. The end of the magnet at which the field lines emerge is designated the north pole (N), and the other end is the south pole (S):

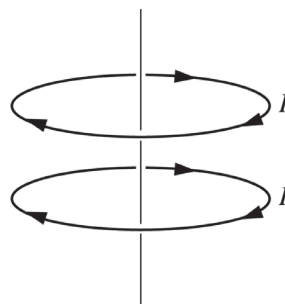
- (a) The figure below shows a bar magnet moving down, through a circular loop of wire. What will be the direction of the induced current in the wire?
- (b) What will be the direction of the induced current in the wire if the magnet is moved as shown in the following diagram?



Exercise

2014_Intl_MCQ_20

Two conducting loops that are centered on the same axis carry equal currents I in the same direction as shown in the diagram above. If the current in the upper loop suddenly decreases to zero, what happens to the current in the lower loop according to Lenz's law?



- (A) It also decreases to zero.
- (B) It decreases, but not to zero.
- (C) It does not change.
- (D) It increases.
- (E) Its direction is reversed.

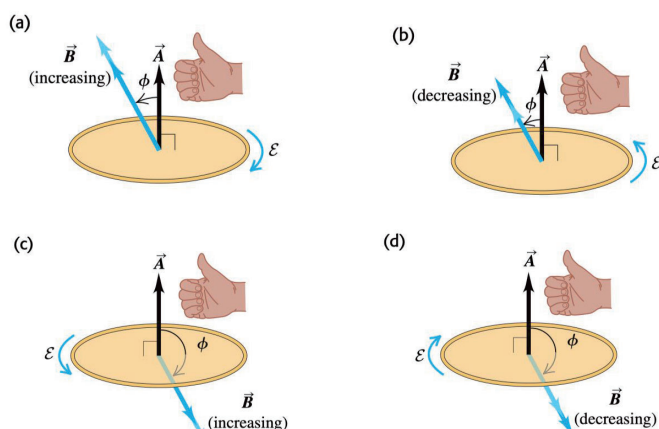
5.1.4

Concept

Sign Convention

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

- Right hand rule



5.1.4

5.1 Electromagnetic induction

5.1.5 Induced Electric Fields

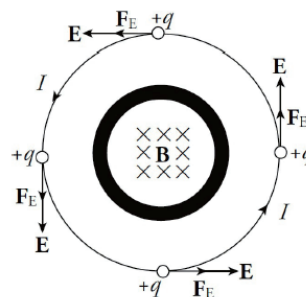
5.1.5

Concept

TESTDAILY

Induced Electric Fields

- What was the source of the force that pushed these charges around the stationary conducting loop?
- There has to be an **induced electric field** in the conductor caused by the changing magnetic flux.



Concept

TESTDAILY

Induced Electric Fields

- The work done on a charge q

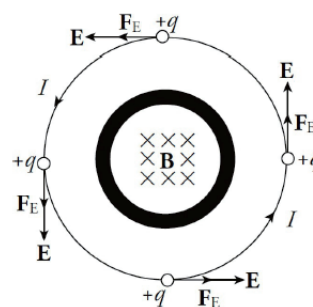
$$W = \oint \vec{F} \cdot d\vec{l} = \oint q\vec{E} \cdot d\vec{l}$$

- Induced emf

$$\varepsilon = \frac{W}{q} = \oint \vec{E} \cdot d\vec{l}$$

- Faraday's law restated: induced electric field and magnetic flux

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$



$$E \cdot 2\pi r = \left| \frac{dB}{dt} \right| A$$



Example

TESTDAILY

Princeton_2020_Chapter16_Example7

For the situation just described, assume that the solenoid has 15,000 turns per meter and a radius of $r = 2$ cm. The radius of the circular loop is $R = 4$ cm. If the current in the solenoid is increased at a rate of 10 A/s, what is the magnitude of the induced electric field at each position along the circular wire?

Answer: 9.4×10^{-4} V/m



Induced Electric Fields

- The induced electric field provides the non-electrostatic force that is the source of the emf.
- Electrostatic fields are conservative, but induced electric fields are not conservative.
- There is no potential or potential energy associated with this vortex electric field.

5.1.5

Inductance

5.2



TESTDAILY

Inductance

5.2.1 — Inductance

5.2.2 — Energy Stored in an Inductor

5.2.3 — Inductors in Series and in Parallel

5.2 Inductance

5.2.1 Inductance

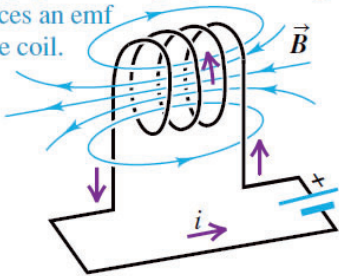


Concept

Self-Inductance

- A solenoid is also called an inductor.
- The very nature of the inductor is to **oppose** the change in current occurring in the inductor.

Self-inductance: If the current i in the coil is changing, the changing flux through the coil induces an emf in the coil.



Concept

Inductance (CED p97)

By Faraday's law, $\varepsilon = -N \frac{d\Phi_B}{dt}$. So, if Φ_B is proportional to current I , then

$$\varepsilon = -L \frac{dI}{dt}$$

where L is defined as the inductance of the electrical device. The self-induced emf is proportional to the **rate of change** of the current.

Physical quantity	Inductance 电感
Definition	The self-induced emf per unit rate of change of current.
Symbol	L
Equation	$\varepsilon = -L \frac{dI}{dt}$
Unit	henry, H $1 \text{ H} = 1 \text{ V}\cdot\text{s}/\text{A} = 1 \Omega\cdot\text{s}$



Concept

TESTDAILY

Alternate Definition of Inductance



Assume that the inductor contains N turns and that Φ_B is the magnetic flux through each turn. Then the total magnetic flux through the entire coil is $N\Phi_B$; this is proportional to the current, so $N\Phi_B = LI$ for some constant, L . The proportionality constant L is called the inductance (or self-inductance) of the coil:

$$L = \frac{N\Phi_B}{I}$$

5.2.1

Example

TESTDAILY

Princeton_2020_Chapter16_Example8

An ideal solenoid of cross-sectional area A and length ℓ contains n turns per unit length. What is its self-inductance?

$$L = \mu_0 n^2 A \ell$$

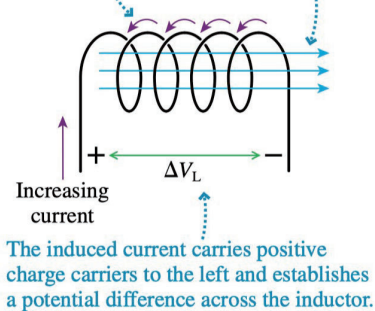
5.2.1

Concept

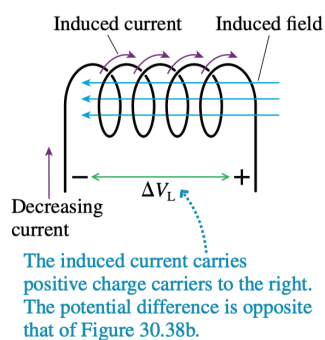
TESTDAILY

Potential Change across an Inductor

The induced current is opposite the solenoid current. The induced magnetic field opposes the change in flux.



5.2.1



5.2 Inductance

5.2.2 Energy Stored in an Inductor



Concept

Energy Stored in an Inductor

$$P = i|\Delta V| = Li\frac{di}{dt}$$

The energy dU supplied to the inductor during an infinitesimal time interval dt is $dU = Pdt$, so

$$dU = Lidi$$

The total energy U supplied while the current increases from zero to a final value I is

$$U = L\int_0^I idi = \frac{1}{2}LI^2$$

This energy is stored in the magnetic fields within the inductor.



5.2 Inductance

5.2.3 Inductors in Series and in Parallel



Concept

TESTDAILY

Equivalent Inductance

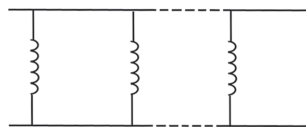


$$I = I_1 = I_2 = \dots$$

$$V = V_1 + V_2 + \dots$$

$$L_s = \sum_i L_i$$

5.2.3

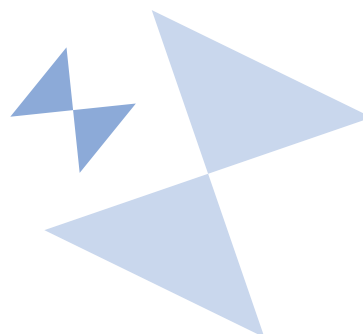


$$V = V_1 = V_2 = \dots$$

$$I = I_1 + I_2 + \dots$$

$$\frac{1}{L_p} = \sum_i \frac{1}{L_i}$$

RL and LC Circuits



TESTDAILY

5.3

RL and LC Circuits

5.3.1 — RL Circuit – Current Growth

5.3.2 — RL Circuit – Current Decay

5.3.3 — LC Circuit

TESTDAILY

5.3 RL and LC Circuits

5.3.1 RL Circuit – Current Growth



Concept

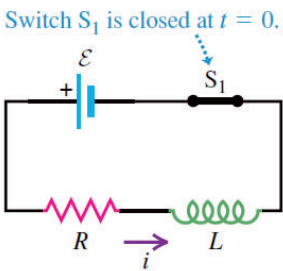
RL Circuit – Current Growth

	$i(t)$	$ \Delta V_R $	$ \Delta V_L $	$\frac{di}{dt}$
$t = 0$				
$t > 0$				
$t = \infty$				

At initial state, the inductor acts like an “open switch”.
At steady-state, the inductor acts like a “wire”.



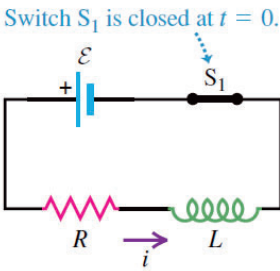
TESTDAILY



Concept

RL Circuit – Current Growth

- Kirchhoff’s loop rule $\varepsilon - iR - L \frac{di}{dt} = 0$
- Solve the differential equation



TESTDAILY

Concept

TESTDAILY

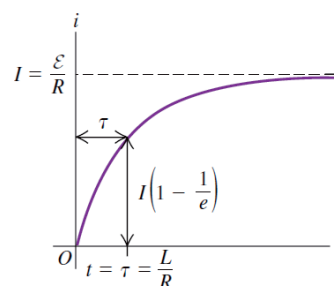
RL Circuit – Current Growth

$$i = \frac{\mathcal{E}}{R} \left(1 - e^{-\frac{t}{L/R}} \right)$$

$$\frac{di}{dt} = \frac{\mathcal{E}}{L} e^{-\frac{t}{L/R}}$$

The quantity L/R is therefore a measure of how quickly the current builds toward its final value; this quantity is called the **time constant** for the circuit, denoted by τ :

$$\tau = \frac{L}{R}$$



5.3.1

Exercise

TESTDAILY

2012_Intl_MCQ_30,31

A circuit consists of a resistor R , an inductor L , and an open switch S connected in series with a battery. The switch is then closed at time $t = 0$.

30. Which of the following quantities could be represented as a function of time by the graph shown above?

- I. The potential difference across the resistor
- II. The potential difference across the inductor
- III. The current in the circuit

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only
- (E) I, II, and III



5.3.1

Exercise

TESTDAILY

31. The change in current when the switch is closed is determined by the inductive time constant τ .

If the inductance is doubled and the resistance is halved, the new inductive time constant τ' equals

- (A) $\frac{1}{4}\tau$
- (B) $\frac{1}{2}\tau$
- (C) τ
- (D) 2τ
- (E) 4τ

5.3.1

5.3 RL and LC Circuits

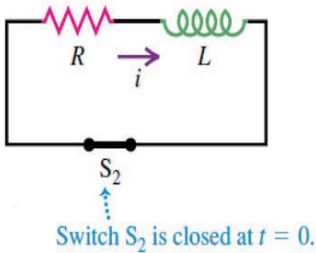
5.3.2 RL Circuit – Current Decay



Concept

RL Circuit – Current Decay

	$i(t)$	$ \Delta V_R $	$ \Delta V_L $	$\frac{di}{dt}$
$t = 0$				
$t > 0$				
$t = \infty$				



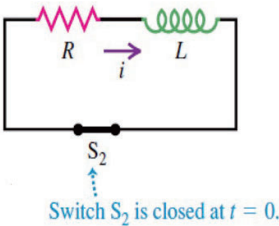
At initial state, the inductor acts like a “battery”.
At steady-state, the inductor acts like a “wire”.



Concept

RL Circuit – Current Decay

- Kirchhoff’s loop rule $-iR - L \frac{di}{dt} = 0$
- Solve the differential equation



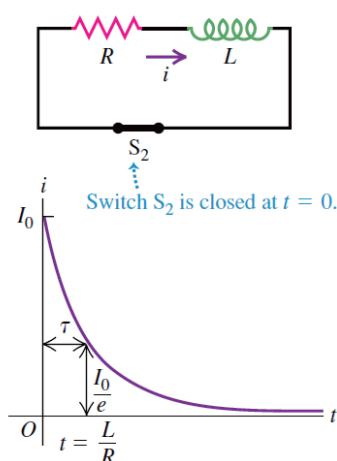
Concept

TESTDAILY

RL Circuit – Current Decay

$$i = I_0 e^{-t/L/R}$$

$$\frac{di}{dt} = -\frac{I_0 R}{L} e^{-t/L/R}$$



5.3.2

Property

TESTDAILY

RL Circuit – Current Decay

- When the current in an R-L circuit is decaying, what fraction of the original energy stored in the inductor has been dissipated after 2.3 time constants?

5.3.2

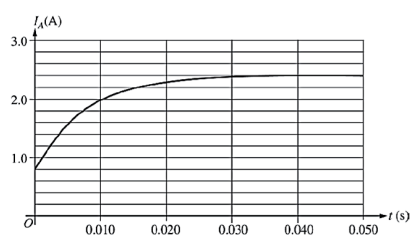
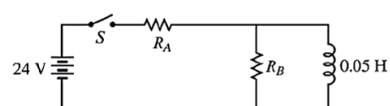
Example

TESTDAILY

2013_Intl_FRQ-1

Students are given the circuit above in their lab, but they do not know the values of the two resistances R_A and R_B . The battery has a terminal voltage of 24 V and the inductor has an inductance of 0.05 H. Assume the internal resistance of both the battery and the inductor to be negligible. The switch S is closed at time $t = 0$. In order to determine the values of the resistances R_A and R_B , the students measure the current as a function of time through the resistance R_A . The graph of their results is below.

- (a) On the circuit diagram above, insert an ammeter to measure the current through R_A .

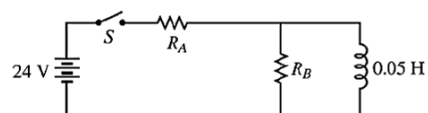


5.3.2

Example

TESTDAILY

- (b) Use the data taken by the students to calculate the value of R_A .
- (c) Use the data taken by the students to calculate the value of R_B .
- (d) Calculate the initial rate of change of current in the inductor.



5.3.2

Example

TESTDAILY

- (e) In reality, both the battery and the inductor have nonnegligible internal resistance. Would the true value of R_A be greater than or less than the value you obtained in part (b)?

_____ Greater than _____ Less than

Justify your answer.

5.3.2

5.3 RL and LC Circuits

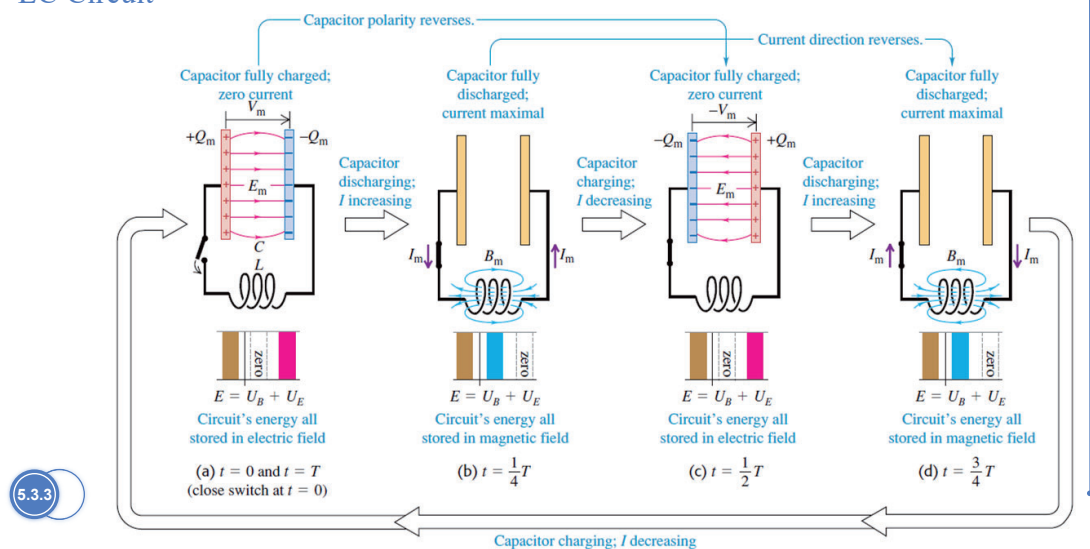
5.3.3 LC Circuit

5.3.3

TESTDAILY

Concept

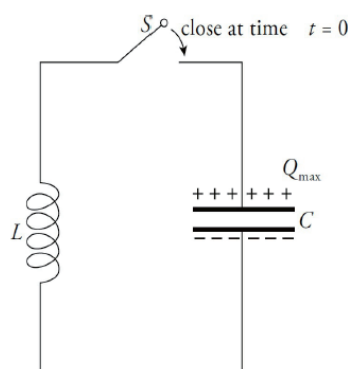
LC Circuit



Property

LC Circuit

- If there are no energy losses, the charges on the capacitor continue to oscillate back and forth indefinitely. This process is called an **electrical oscillation**.



Property

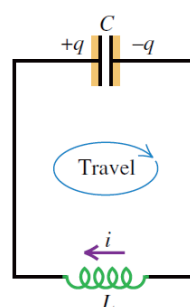
LC Circuit

We apply Kirchhoff's loop rule and obtain

$$-L \frac{di}{dt} - \frac{q}{C} = 0$$

Since $i = dq/dt$, it follows that $di/dt = d^2q/dt^2$.

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$



Property

TESTDAILY

LC Circuit

- Charge and current

$$q = Q \cos(\omega t + \phi)$$

$$i = -\omega Q \sin(\omega t + \phi)$$

- Angular frequency

$$\omega = \sqrt{\frac{1}{LC}}$$

- Period

$$T = 2\pi\sqrt{LC}$$



Property

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LC Circuit

The LC circuit is also a conservative system. The sum of magnetic-field energy and electric-field energy equals the total energy of the system

$$U_B + U_E = U_{total}$$

$$\frac{1}{2}Li^2 + \frac{q^2}{2C} = \frac{Q^2}{2C}$$

$$i = \pm \sqrt{\frac{1}{LC}} \sqrt{Q^2 - q^2}$$



Example

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Young_Chapter30_Example30.8

A 300-V DC power supply is used to charge a 25- μ F capacitor. After the capacitor is fully charged, it is disconnected from the power supply and connected across a 10-mH inductor. The resistance in the circuit is negligible.

- Find the frequency and period of oscillation of the circuit.
- Find the capacitor charge and the circuit current 1.2 ms after the inductor and capacitor are connected.

Answer: 320 Hz, 3.1 ms, -5.5×10^{-3} C, -10 A



Example

TESTDAILY

Young_Chapter30_Example30.9

For the L - C circuit of Example 30.8, find the magnetic and electric energies

- (a) at $t = 0$ and
 (b) at $t = 1.2$ ms.

Answer: 0, 1.1 J; 0.5 J, 0.6 J.

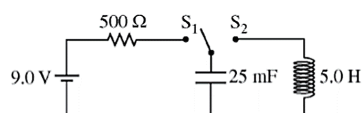
5.3.3

Example

TESTDAILY

2011_NA_FRQ_2 (b)

The circuit represented above contains a 9.0 V battery, a 25 mF capacitor, a 5.0 H inductor, a 500 Ω resistor, and a switch with two positions, S_1 and S_2 . Initially the capacitor is uncharged and the switch is open.



(b) In experiment 2 the capacitor is again uncharged when the switch is closed to position S_1 at time t_1 . The switch is then moved to position S_2 at time t_2 when the magnitude of the charge on the capacitor plate is 105 mC, allowing electromagnetic oscillations in the LC circuit.

- i. Calculate the energy stored in the capacitor at time t_2 .

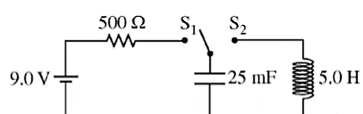
5.3.3

Example

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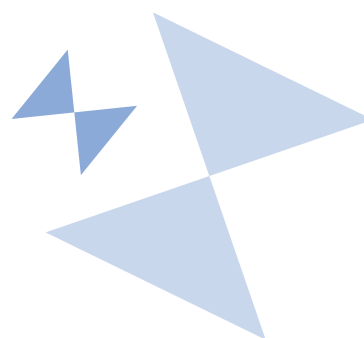
2011_NA_FRQ_2 (b)

- ii. Calculate the maximum current that will be present during the oscillations.
 iii. Calculate the time rate of change of the current when the charge on the capacitor plate is 50 mC.



5.3.3

Maxwell's Equations



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5.4

Concept

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Maxwell's Equations

Gauss's Law for electricity

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

Gauss's law for magnetism

$$\oint \vec{B} \cdot d\vec{A} = 0$$

Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}}$$

Faraday's law

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$

5.4

Concept

TESTDAILY

Gauss's Law for Electricity

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

5.4

Concept

TESTDAILY

Gauss's Law for Magnetism

$$\oint \vec{B} \cdot d\vec{A} = 0$$

- There is no magnetic monopole.
- Magnetic field lines are closed loops.

5.4

Concept

TESTDAILY

Faraday's Law

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

5.4

Concept

TESTDAILY

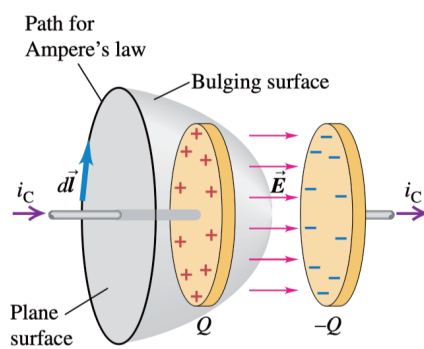
The Ampere-Maxwell Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}}$$

Define displacement current:

$$i_D = \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_C + i_D)_{\text{encl}}$$



5.4

Example

TESTDAILY

Princeton_2020_Chapter16_Example10

Which one of Maxwell's equations states that a changing electric field produces a magnetic field?

- (A) Gauss's law
- (B) Gauss's law for magnetism
- (C) Biot-Savart law
- (D) Ampere-Maxwell law
- (E) Faraday's law

5.4

Answer to MCQ

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Section	Source	Answer
5.1.2	2013_Intl_MCQ_6	A
5.1.3	2016_Intl_MCQ_22	E
5.1.3	2021_DE_D61102_11	E
5.1.4	2014_Intl_MCQ_20	D
5.3.1	2012_Intl_MCQ_30,31	C, E

5

Answer to FRQ

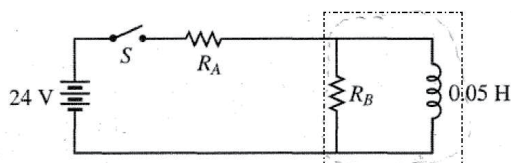
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2013_Intl_FRQ-1

15 points total

Distribution of points

- (a) 1 points



For inserting an ammeter in series anywhere in the circuit above EXCEPT inside the dashed box

1 point

5

Answer to FRQ

TESTDAILY

2013_Intl_FRQ-1

(b) 3 points

At steady state, the potential difference across the inductor and R_B is zero. So R_A can be determined using the 24 V battery potential difference and the steady-state current from the graph.

For a correct use of Ohm's law

$$V = IR_A$$

For correct substitutions into the above equation

$$R_A = \frac{V}{I} = \frac{(24 \text{ V})}{(2.4 \text{ A})}$$

For a correct answer

$$R_A = 10 \, \Omega$$

If a student did not use 2.4 A, the substitution point was lost if the value of I was in the range: $2.2 < I < 2.6$. If the value was outside this range, both the substitution and the answer point were lost.

5

1 point

1 point

1 point

Answer to FRQ

TESTDAILY

2013_Intl_FRQ-1

(c) 4 points

For an indication that there is no current through the inductor at time $t = 0$

For correct use of Ohm's law with the resistors in series

$$V = I(R_A + R_B)$$

$$V/I = (R_A + R_B)$$

$$R_B = V/I - R_A$$

For correct substitution of the correct initial value of I from the graph and the resistance from part (b)

$$R_B = \frac{(24 \text{ V})}{(0.8 \text{ A})} - (10 \, \Omega)$$

For an answer consistent with part (b)

$$R_B = 20 \, \Omega$$

5

1 point

1 point

1 point

1 point

Answer to FRQ

TESTDAILY

2013_Intl_FRQ-1

(d) 5 points

The voltage across the inductor is equal to the voltage across resistor R_B

For using the correct voltage across R_B

$$V_B = I_{Ai} R_B$$

For equating $L \frac{dI}{dt}$ and V_B

$$L \frac{dI}{dt} = I_{Ai} R_B$$

Solve for dI/dt and substitute correct values

$$\frac{dI}{dt} = \frac{I_{Ai} R_B}{L} = \frac{(0.8 \text{ A})(20 \, \Omega)}{(0.05 \text{ H})}$$

For a correct answer consistent with part (c)

For correct units

$$\frac{dI}{dt} = 320 \text{ A/s}$$

5

2 points

1 point

1 point

1 point

Answer to FRQ

TESTDAILY

2013_Intl_FRQ-1

(e) 2 points

For selecting "Less than"

1 point

For a correct justification

1 point

Example: Since R_A is in series with the battery, then the $10\ \Omega$ calculated value would equal $R_A + R_{\text{battery}}$, therefore the true value would be $R_A = 10\ \Omega - R_{\text{battery}}$, which is less.



Answer to FRQ

TESTDAILY

2011_NA_FRQ_2 (b)

(b)

i. 2 points

$$U_C = \frac{1}{2} QV = \frac{1}{2} Q \frac{Q}{C} = \frac{1}{2} \frac{Q^2}{C}$$

For substituting correct values into a correct expression

1 point

$$\text{For example, } U_C = \frac{1}{2} \frac{(105 \times 10^{-3} \text{ C})^2}{(25 \times 10^{-3} \text{ F})}$$

For a consistent answer with correct units

1 point

$$U_C = 0.22 \text{ J}$$



Answer to FRQ

TESTDAILY

2011_NA_FRQ_2 (b)

ii. 2 points

The maximum current occurs when there is no charge on the capacitor and all the energy is stored in the inductor.

$$U_L = \frac{1}{2} LI^2$$

The total energy is the energy that was stored in the capacitor at time t_2 .

For a correct expression of energy conservation

1 point

$$\frac{1}{2} LI^2 = U_C$$

$$I = \sqrt{2U_C/L}$$

Substituting the given value for L and the value of U_C determined in part (b) i

$$I = \sqrt{2(0.22 \text{ J})/5.0 \text{ H}}$$

For an answer with units consistent with previous work

1 point

$$I = 0.30 \text{ A}$$



Answer to FRQ

TESTDAILY

2011_NA_FRQ_2 (b)

iii. 3 points

For a correct application of the loop rule

1 point

$$L \frac{dI}{dt} + \frac{Q}{C} = 0$$

$$\frac{dI}{dt} = -\frac{Q}{CL}$$

$$\frac{dI}{dt} = -\frac{(50 \times 10^{-3} \text{ C})}{(25 \times 10^{-3} \text{ F})(5.0 \text{ H})}$$

For a correct numerical answer obtained from a correct procedure, with or without the negative sign

1 point

For the correct units on a calculated answer

1 point

$$\frac{dI}{dt} = -0.40 \text{ A/s}$$

5

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5

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5

